

Theory

Semester II

45 contact hours

3 credits

CIE 40 / ESE 60

2182 Building Materials & Technology

Conventional, modern and sustainable building materials with essential construction technologies for building components and finishes.

Course code	2182
Scheme	2:1:0:0 printed course table
Credits	3
Contact hours	45
Self-learning	4 / week; 60 h listed
Type	Theory

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official syllabus PDF was used as the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, instructional hours, self-learning activities, assessment structure and listed learning resources. Educational explanations, examples and diagrams have been added to make the official topics learnable; they are not presented as additional official syllabus requirements.

Source file: 2182.pdf

Curriculum: Diploma Curriculum Revision 2026.

The course-information table prints teaching scheme 2:1:0:0, while the printed Teaching Scheme/Week row shows 3:0:0:0. Both values are preserved here; the source PDF is not silently corrected.

AT A GLANCE

Course dashboard

Course dashboard	
Field	Official information
Programme	Architecture; Interior Design
Course code / title	2182 — Building Materials & Technology
Semester / type	II / Theory
Teaching scheme	2:1:0:0 in course table; printed assessment row shows 3:0:0:0
Contact hours / credits	45 / 3
Self-learning	4 per week; total 60 hours
Assessment	CIE 40; ESE 60
Prerequisite	Mechanical and chemical properties of materials; Secondary Education

STUDY METHOD

How to use this lesson

Before class

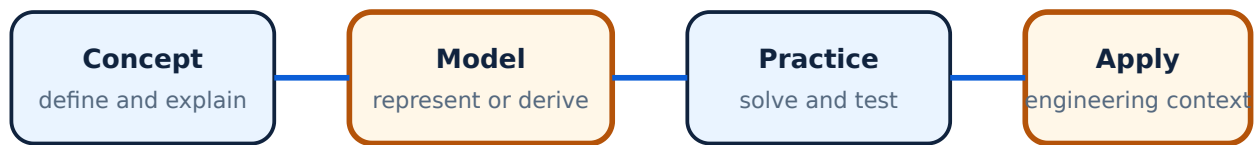
Read the topic definition and inspect the diagram. Mark unfamiliar technical words in the glossary.

During practice

Reproduce the worked example or procedure without looking at the answer. For theory topics, redraw labelled diagrams and compare classifications.

Before examination

Use the module questions, formula bank, official assessment pattern and final checklist. Treat the practice paper as preparation, not as an official question paper.



Use this cycle for every module: understand the official concept, represent it accurately, practise, and connect it to an application.

OUTCOMES AND ALIGNMENT

Objectives, course outcomes and CO-PO mapping

Official course objectives

1. Understand classification, properties and sustainable aspects of common building materials.
2. Gain knowledge of concrete materials, mix proportions, construction processes and finishes.
3. Comprehend building components, structural elements and foundation systems.
4. Become familiar with masonry, roofing, formwork, scaffolding and modern construction technologies.

Course outcomes		
CO	Official course outcome	Bloom
CO1	Classify building materials and explain properties, defects and sustainability of stone, timber, bamboo and eco-friendly masonry blocks.	Understand
CO2	Explain cement, concrete ingredients, mix proportions, grades, concreting, curing, defects and flooring/finishing materials.	Understand

CO	Official course outcome	Bloom
CO3	Identify building components and distinguish load-bearing/non-load-bearing elements, foundations and structural systems.	Understand
CO4	Apply knowledge of masonry, bricks and bonds, formwork, scaffolding, roofing and vertical circulation.	Apply

CO-PO course articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	1	-	-	1	-	1
CO2	2	3	2	-	2	3	1	-	1	-	1
CO3	3	3	2	2	2	1	-	-	1	-	1
CO4	2	2	3	2	1	1	1	2	2	1	1

Legend: 1 = Low; 2 = Medium; 3 = High; - = No Correlation.

NOTHING OMITTED

Complete syllabus coverage

Complete syllabus coverage map

Module	Official title	Instructional hours	Major syllabus coverage
I	Fundamentals of Building Materials	7 L + 4 T	Classification; stone, timber, bamboo; clay, fly-ash, AAC and CSEB blocks
II	Concrete Materials and Building Finishes	7 L + 4 T	Cement, concrete, grades, water-cement ratio, processes, curing, defects, flooring, finishes, composites, metals, insulation and paints
III	Building Components & Foundation Systems	8 L + 4 T	Components, structural systems, building elements and shallow/deep foundations
IV	Masonry, Roofing & Construction Technology	7 L + 4 T	Masonry, bricks/bonds, formwork, scaffolding, vertical circulation and roof forms

Every official module and major topic is expanded below. The wording may be grouped for teaching clarity, but no supplied syllabus item is intentionally removed.

Official roof-form term: Roof classification includes the *flat roof* in addition to pitched, curved and

trussed roofs. Compare each by form, span, drainage, materials and structural support.

LEARNING ROADMAP

How the modules connect

Module 1
Materials

Module 2
Concrete & finishes

Module 3
Components & foundations

Module 4
Masonry & roofs

MODULE 1

Fundamentals of Building Materials

Material selection begins with origin, manufacture, properties, defects, service conditions and sustainability.

Official hours: 7 L + 4 T

Mapped CO: CO1

Bloom level: Understand

Learning goals

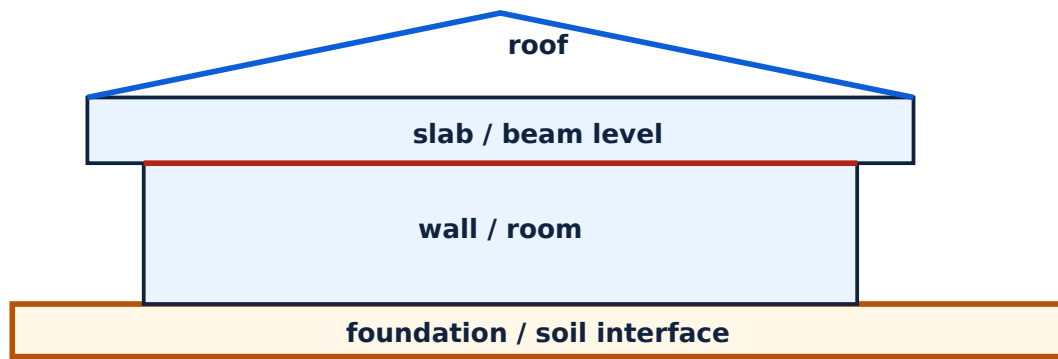
- Classify materials as natural, artificial, engineered and green.
- Compare stone, timber and bamboo.
- Recognize eco-friendly masonry blocks and their uses.

Key facts

- Seasoning reduces timber moisture before use.
- A defect is a departure from required quality.
- Green materials should be assessed by life-cycle impact, not a label alone.

Engineering relevance

Use the concepts to interpret technical information, communicate decisions accurately, and solve appropriately scoped problems.



Building components are organized as a load-transfer and enclosure system from roof to foundation, while openings and finishes meet functional requirements.

Classification of building materials–

Natural materials occur with limited processing, artificial materials are manufactured from constituents, engineered materials are designed for a target performance, and green materials emphasize reduced environmental impact or resource use. A classification chart should include origin, processing, properties and example applications.

Malayalam support: Material classify നാറ്റുറൽ നാച്ചുറൽ മാനുഫാക്ചർഡ് പ്രോപ്പർട്ടീസ് പ്രൊസസ്സിംഗ് പ്രൊഫർമൻസ്, ഡ്യൂറബിലിറ്റി, സസ്റ്റൈനബിലിറ്റി ഉദാഹരണങ്ങൾ.

Stone: types, properties, defects and uses–

Stone selection considers strength, durability, porosity, texture, weathering resistance and appearance. Common defects may arise from cracks, laminations, cavities or weathering. Use stone appropriately in foundations, walls, flooring, cladding or landscape work based on performance.

Timber: properties, defects, seasoning and uses–

Timber is valued for workability, strength-to-weight ratio and appearance. Defects include knots, shakes, checks, warping and decay. Seasoning reduces moisture to improve dimensional stability and service performance. Protect it from moisture, fungi, insects and fire as required.

Bamboo and eco-friendly masonry blocks–

Bamboo is renewable and can be used in structural, enclosure and finishing applications when properly selected, treated and detailed. Compare clay bricks, fly ash bricks, AAC blocks and CSEB

by density, strength, thermal behavior, water absorption, manufacturing and appropriate applications.

Eco-friendly masonry block comparison

Block	Useful comparison points
Clay brick	Traditional masonry unit; compare strength, absorption and firing energy.
Fly ash brick	Uses industrial by-product; consistent shape when properly manufactured.
AAC block	Lightweight cellular unit with thermal advantages; handle carefully.
CSEB	Compressed stabilized earth block; uses local earth with stabilizer and appropriate protection.

MODULE 2

Concrete Materials and Building Finishes

Concrete performance depends on constituent quality, proportions, water-cement ratio, placement, compaction and curing; finishes complete the building surface.

Official hours: 7 L + 4 T

Mapped CO: CO2

Bloom level: Understand

Learning goals

- Differentiate OPC, rapid-hardening, quick-setting and white cement.
- Explain ingredients, PCC/RCC, grades and process.
- Select flooring, finishing, composite, metal, insulation and coating materials.

Key facts

- Water-cement ratio influences workability and strength.
- Compaction reduces voids.
- Curing maintains moisture and supports hydration.

Engineering relevance

Use the concepts to interpret technical information, communicate decisions accurately, and solve appropriately scoped problems.

Cement types and concrete ingredients–

OPC is a general-purpose cement; rapid-hardening cement develops strength faster; quick-setting cement sets quickly; white cement is selected where colour is important. Concrete combines

cement, water, fine aggregate and coarse aggregate. Each ingredient influences workability, strength, durability and finish.

PCC, RCC, grades, mix proportions and water-cement ratio–

PCC is plain cement concrete without reinforcing steel; RCC includes reinforcement to resist tensile effects in a composite system. The syllabus lists M15, M20 and M25 grades and simple mix proportions. Do not confuse a grade designation with a complete structural mix design.

$w/c = \text{water mass} \div \text{cement mass}.$

Given: 25 kg water and 50 kg cement.

Required: w/c ratio.

Calculation: $w/c = 25/50.$

Answer: $w/c = 0.50.$

Mixing, placing, compaction, curing and defects–

Concreting proceeds through batching/mixing, transporting, placing, compaction and finishing, followed by curing. Compaction removes entrapped air. Curing preserves moisture and temperature conditions for hydration. Defects can include honeycombing, cracks, segregation and bleeding; identify causes and preventive precautions.

Flooring, finishing, composites, metals, insulation and coatings–

Stone, tile, cement and terrazzo flooring are selected by use, wear, moisture, appearance and maintenance. Plaster, putty and surface finishes prepare/protect surfaces. Plywood, MDF and particle board are composites with different fibre/particle construction. Steel and aluminium serve different structural, enclosure and corrosion contexts. Thermal and acoustic insulation, coatings and paints address comfort, protection and appearance.

MODULE 3

Building Components & Foundation Systems

A building is read as a coordinated set of substructure, superstructure, structural elements, openings, circulation and finishes.

Learning goals

- Identify substructure and superstructure.
- Differentiate load-bearing and framed systems.
- Explain walls, columns, beams, slabs and building elements.
- Classify shallow and deep foundations.

Key facts

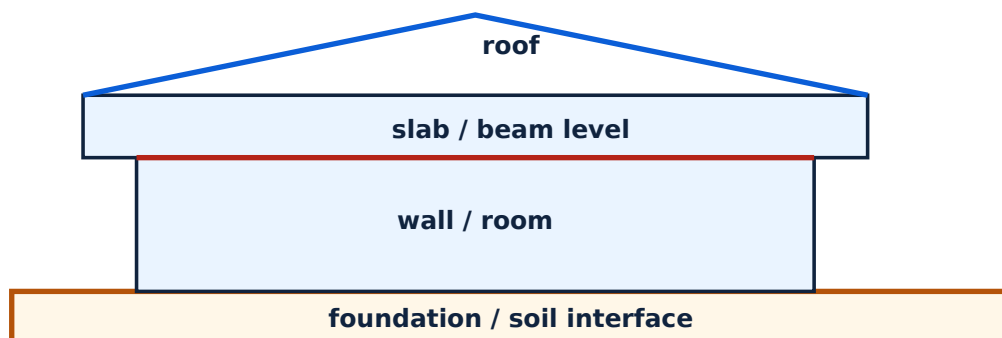
- A foundation transfers load to soil.
- Openings must meet functional requirements.
- Foundation choice depends on load and soil conditions.

Engineering relevance

Use the concepts to interpret technical information, communicate decisions accurately, and solve appropriately scoped problems.

Basic components: substructure and superstructure–

Substructure lies below or near ground level and includes foundations; superstructure rises above it and includes walls, columns, beams, slabs, roofs and enclosures. Draw a labelled section to show the relationship rather than presenting isolated definitions.



Building components are organized as a load-transfer and enclosure system from roof to foundation, while openings and finishes meet functional requirements.

Load-bearing and non-load-bearing systems–

In a load-bearing system, walls carry significant vertical loads. In a framed system, columns and beams form the main load path and partitions may be non-load-bearing. Compare load path, flexibility, openings, construction sequence and suitability without assuming one system is universally better.

Walls, columns, beams, slabs, doors, windows, staircases, roofs and finishes–

Walls enclose or divide; columns carry axial and combined actions; beams transfer loads; slabs form floors/roofs. Doors provide controlled access, windows provide light/ventilation/views, staircases provide vertical movement, roofs shelter and finishes protect/complete surfaces. State functional requirements when describing each element.

Shallow and deep foundations–

Shallow foundations distribute loads near the surface: strip, isolated, combined footing and raft are listed. Deep foundations transfer load to deeper strata or develop resistance through piles or piers when near-surface soil or loading conditions require it. Selection is a site and design decision, not a visual preference.

MODULE 4

Masonry, Roofing & Construction Technology

Construction technology connects material properties with repeatable site methods, safe temporary works, circulation and roof form.

Official hours: 7 L + 4 T

Mapped CO: CO4

Bloom level: Apply

Learning goals

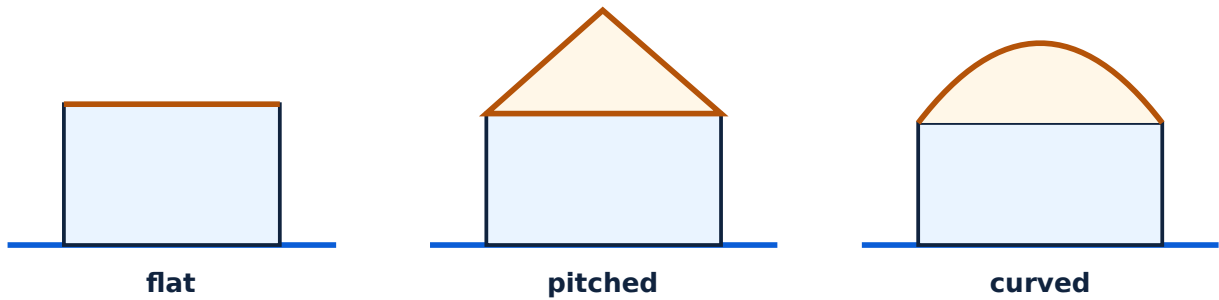
- Differentiate stone, brick and block masonry.
- Recall brick dimensions/properties and identify bonds.
- Explain formwork and scaffolding.
- Select basic roof forms and vertical-circulation systems.

Key facts

- Bond controls joint arrangement and continuity.
- Formwork is temporary but safety-critical.
- Roof selection depends on form, material, climate, span and drainage.

Engineering relevance

Use the concepts to interpret technical information, communicate decisions accurately, and solve appropriately scoped problems.



Roof classification considers form, material and structural system. Selection depends on climate, span, drainage, construction and use.

Masonry, bricks and brick bonds–

Stone, brick and block masonry differ in unit shape, jointing, material and construction technique. The syllabus requires standard brick sizes, properties and characteristics, plus English, Flemish, rat-trap, header and stretcher bonds. Draw bonds with clear bed and vertical joints.

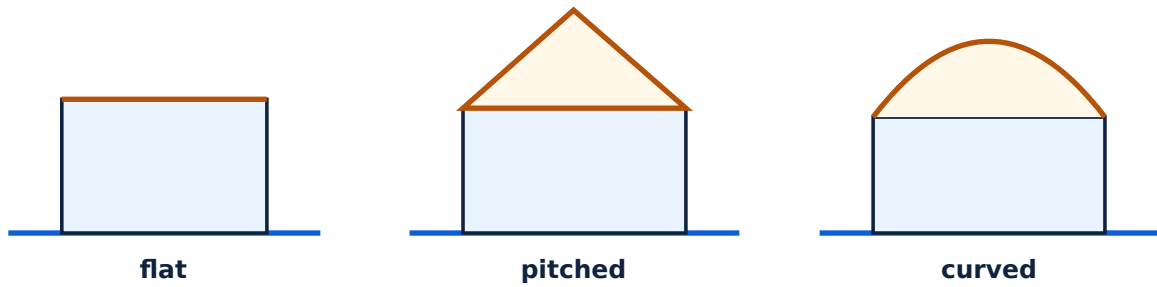
Formwork and scaffolding–

Formwork supports fresh concrete until it gains adequate strength. Materials may include timber, steel or proprietary systems; compare reuse, accuracy, cost, erection and stripping. Scaffolding provides temporary access and working platforms; describe purpose and types and include safe loading, bracing and access principles.

Safety note: Temporary works must be designed, inspected and used under competent supervision; never treat a sketch as a construction approval.

Vertical circulation and roofing systems–

Lifts, stairs and escalators serve different movement patterns, capacities and accessibility needs. Roofs may be classified by shape, material and structural system. Flat, pitched, curved and trussed roofs require different drainage, span, covering and support considerations.



Roof classification considers form, material and structural system. Selection depends on climate, span, drainage, construction and use.

RAPID CALCULATION REFERENCE

Formula bank

Water-cement ratio

$$w/c = \text{mass of water} \div \text{mass of cement}$$

Use consistent mass units; the ratio affects workability and strength.

Concrete grade

M20 indicates characteristic compressive-strength class conventionally associated with 20 N/mm² at the specified age

The syllabus lists M15, M20 and M25; do not treat the grade as a mix-proportion recipe by itself.

Area of rectangular slab

$$A = L \times B$$

Use consistent length units and report square units.

Load path

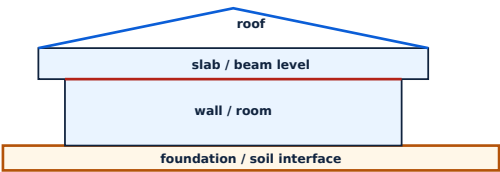
Roof → slab/beam → wall/column → foundation → soil

This is a conceptual structural sequence; actual design requires qualified engineering analysis.

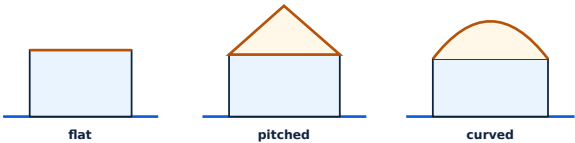
RAPID VISUAL REVISION

Diagram library

Use these labelled schematics to revise relationships and redraw them in examination answers where relevant.



Building components are organized as a load-transfer and enclosure system from roof to foundation, while openings and finishes meet functional requirements.



Roof classification considers form, material and structural system. Selection depends on climate, span, drainage, construction and use.

OFFICIAL SELF-LEARNING

Activities and independent study

Suggested student activities for self-learning — wording retained in substance from the source PDF		
Week / group	Official self-learning activity	Hours
1-3	Classification chart; stone/timber/bamboo case study; eco-friendly block comparison.	12
4-8	Cement comparison; concrete ingredient/water-cement flowchart; grades/PCC/RCC note; concreting/curing documentation; materials identification.	20
9-12	Building-component diagrams; load-bearing comparison; component sketches; foundation case study.	16
13-15	Masonry/brick documentation; bonds/formwork/scaffolding illustration; roofing and vertical-circulation chart.	12

Submission tip: Keep a dated record of sketches, calculations, code/output screenshots, case-study evidence or field observations as appropriate. Use the exact activity title from the syllabus when submitting.

EXAMINATION READINESS

Assessment methodology

Theory assessment structure shown in the supplied Revision 2026 syllabus PDF.		
Component	Official mode / duration	Marks / timing
Attendance	Attendance; end of semester	5 converted marks

Component	Official mode / duration	Marks / timing
CA1	Self-learning activities, Module 1; by week 4	15
CA2	Self-learning activities, Module 2; by week 7	15
CA3	Self-learning activities, Module 3; by week 11	15
CA4	Self-learning activities, Module 4; by week 14	15
CA5	Series examination, two modules; 2 hours; week 8	50 raw → 20 converted
CA6	Series examination, two modules; 2 hours; week 15	50 raw → 20 converted
ESE	Written theory; 2.5 hours; end of semester	60

Series examination pattern — theory

Part	Question pattern	Marks
Part A	4 questions total; 2 from each selected module; 1 mark each	4
Part B	6 questions total; 3 from each selected module; 3 marks each	18
Part C	4 questions total; 2 with choice from each selected module; 7 marks each	28
Total	Two selected modules, 2 hours	50

End Semester Examination pattern — theory

Module	Part A	Part B	Part C	Module total
1	$2 \times 1 = 2$	2 of $3 \times 3 = 6$	1 set with choice $\times 7$	15
2	2	6	7	15
3	2	6	7	15
4	2	6	7	15
Total	8	24	28	60

Important: The practice pattern below is generated for revision and must not be represented as an official examination paper.

MODULE-WISE QUESTION BANK

Questions and answers

Answers are deliberately matched to the likely mark value: definitions are concise, while longer answers use structure, comparison, formulae or diagrams.

Module I — Materials

What is a green building material?+

State two timber defects.+

Name four eco-friendly masonry blocks in the syllabus.+

Module II — Concrete and finishes

What is PCC?+

What does water-cement ratio mean?+

List the main concreting stages.+

Module III — Components and foundations

Differentiate substructure and superstructure.+

What is a load-bearing structure?+

Name two deep foundation types.+

Module IV — Construction technology

Name two brick bonds.+

Why is formwork used?+

Name four roof forms.+

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Series examination pattern — theory

Part	Question pattern	Marks
Part A	4 questions total; 2 from each selected module; 1 mark each	4
Part B	6 questions total; 3 from each selected module; 3 marks each	18
Part C	4 questions total; 2 with choice from each selected module; 7 marks each	28
Total	Two selected modules, 2 hours	50

End Semester Examination pattern — theory

Module	Part A	Part B	Part C	Module total
1	2 × 1 = 2	2 of 3 × 3 = 6	1 set with choice × 7	15

Module	Part A	Part B	Part C	Module total
2	2	6	7	15
3	2	6	7	15
4	2	6	7	15
Total	8	24	28	60

LAST-MINUTE REVIEW

Quick revision and exam checklist

Module-wise key points

1. **Module 1:** Classify materials; compare stone, timber, bamboo and eco-friendly blocks.
2. **Module 2:** Cement, concrete, w/c ratio, concreting, curing, defects and finishes.
3. **Module 3:** Building components, load paths, structural systems and foundations.
4. **Module 4:** Masonry, brick bonds, formwork, scaffolding, circulation and roof forms.

Exam checklist

- Write definitions before explanations.
- Use standard symbols, units, labels and terminology.
- For comparisons, use a table with a clear basis.
- For numerical work, show Given, Required, Calculation and Answer.
- For diagrams, include a title, labels and a one-line interpretation.
- Check that the answer addresses the command word: define, explain, compare, apply or discuss.

TECHNICAL VOCABULARY

Glossary

Aggregate

Granular material used with cement and water in concrete.

Water-cement ratio

Mass ratio of water to cement.

PCC

Plain cement concrete.

RCC

Reinforced cement concrete.

Foundation

Substructure transferring building loads to soil.

Formwork

Temporary mould/support for fresh concrete.

Scaffolding

Temporary access and working platform system.

AAC

Autoclaved aerated concrete block.

CSEB

Compressed stabilized earth block.

SOURCE-SUPPORTED RESOURCES

Official learning resources

Only resources listed in the supplied syllabus PDF are included here.

- Textbook: *Building Materials and Technology*, SITTTTR Kalamassery.
- *Building Construction*, B. C. Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi Publications.
- *Building Materials*, S. K. Duggal, New Age International.
- *Construction Technology*, Subir K. Sarkar, S. Chand.
- *Building Construction Handbook*, R. Chudley and R. Greeno, Routledge.
- *Materials for Architects and Builders*, Arthur Lyons, Routledge.